NFC. SMART CARDS. SECURE TECHNOLOGY.

Everything NFC, from one partner.

Hardware, software and cloud for payment, identity and connected products.

BANKING & PAYMENT / IDENTITY / ACCESS / BRAND PROTECTION / IoT

COMPANY OVERVIEW

One partner across the complete NFC stack.

NFCTEC develops NFC, smart card and secure technology solutions that connect physical credentials with software and services.

01

Integrated engineering

RF, embedded firmware, card applications and host software developed around the same system requirements.

02

Flexible delivery

Engage us for a reader, an SDK, a card application or an integrated solution with cloud services.

03

Prototype to deployment

Support across architecture, prototyping, integration, testing, personalization and production preparation.

ENGINEERING CAPABILITIES

Technology that connects cards, readers and services.

Practical engineering across the RF interface, card operating system and security layer.

01

NFC & RF

Reader controllers, antennas, contactless protocols and embedded firmware.

ISO/IEC 14443 / 15693
NFC Forum / NDEF
PN5180 / ST25R3916

02

Smart cards

APDU communication, card applications, secure channels, personalization and issuance.

Java Card / GlobalPlatform
SCP02 / SCP03 / EMV
DESFire / MIFARE / FeliCa

03

Security & identity

Key management, secure elements and card-based authentication for identity and wallet applications.

PIV / ePassport / FIDO2
HSM integration / SE050
On-card signing

PRODUCTS & SERVICES

Software. Hardware. Cloud.

Choose individual components or combine them into an integrated solution.

SOFTWARE

SDKs & tools

Reader SDKs for Windows, Linux and Android. Java Card, GlobalPlatform and PC/SC integration. Desktop tools for personalization and testing.

APDU / NDEF / PC/SC

HARDWARE

Readers & modules

Desktop and embedded readers with USB, serial and CCID interfaces. NFC modules and antennas for ESP32, STM32 and other host platforms.

USB / UART / SPI / I2C

CLOUD

Issuance & verification

APIs for credential issuance, verification and status. Key-management service integration to support secure backend operations.

REST / HTTPS / HSM integration

INDUSTRY SOLUTIONS / BANKING & PAYMENT

Bank cards, from data preparation to issuance.

Engineering support for payment applications, card personalization and secure issuance infrastructure.

Concept illustration - bank payment and card issuance applications

Payment integration

EMV APDU workflows, contact and contactless card interfaces, reader and terminal integration.

Data preparation & personalization

Issuer data mapping, application profiles, personalization data generation and validation; card issuance and post-issuance checks.

Key management & HSM integration

Key generation and derivation, secure provisioning and lifecycle workflows, with HSM-backed cryptographic operations.

INDUSTRY SOLUTIONS / 01

Identity and access, connected.

Reader integration, secure credentials and host software for identity and access workflows.

Concept illustration - identity and access applications

Identity verification

ePassport and PIV communication, document-reader integration and authentication interfaces.

Access & transit

DESFire, MIFARE and FeliCa readers, OSDP integration and credential issuance workflows.

INDUSTRY SOLUTIONS / 02

A tap that connects products to verification.

NFC-based brand protection and traceability, from tag personalization to backend verification.

Concept illustration - product authentication application

Secure NFC labels

NTAG 424 DNA integration, tag configuration and SUN message verification.

Connected workflows

Mobile experiences and backend APIs that connect a physical tag with product data and verification results.

INDUSTRY SOLUTIONS / 03

Connect devices through NFC.

Embedded NFC interfaces, firmware and secure identity for industrial and connected products.

Concept illustration - industrial IoT application

Device commissioning

NFC-enabled setup and provisioning workflows integrated with embedded firmware and host applications.

Secure device identity

Secure-element integration and backend interfaces for device credentials and service connectivity.

SECURE WALLET ENGINEERING

Card-based wallets and on-card signing.

Hardware wallets and cold-storage cards extend our work in NFC, smart cards and secure key storage.

Security architecture

Design around a card or secure element that holds the private key and performs signing internally. Integrate the host application, reader and transaction workflow around that boundary.

Card and secure-element integration

Key provisioning, signing interfaces and communication with desktop, mobile or embedded hosts.

Wallet protocols and cryptography

secp256k1 / ed25519
BIP-32 / BIP-39
Implementation depends on the selected chip and application.

PROJECT DELIVERY

A clear path from concept to deployment.

Scope, interfaces and acceptance criteria are agreed before implementation.

01

Define

Requirements, chip selection and system architecture.

02

Build

Prototype hardware, firmware, SDKs and card applications.

03

Integrate

Security, issuance, verification and backend services.

04

Validate

Testing with target cards, readers and operating environments.

Deployment support: personalization, production preparation and firmware updates within the agreed project scope.

LET'S BUILD YOUR NEXT NFC SOLUTION.

nfctec.com

Start with your application.

Tell us what the card needs to do, how the reader connects, and where the service will run.

Discuss a product or engineering project

Cards & readers / Firmware & SDKs / Secure applications / Cloud integration